

Übungsprotokoll

SYTB – Systemtechnik Betriebssysteme

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	Abgabedatum: 16.04.2026	Gruppe: SYTB2	Note:
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Übungsbezeichnung: DHCP & DNS Windows			

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1 Aufgabenstellung

Du sollst einen Windows Server 2025 so konfigurieren, dass er:

- Domänencontroller (DC) wird
- Active Directory (AD DS) bereitstellt
- DNS-Server betreibt
- DHCP-Server konfiguriert

2 Abstract (English)

In this exercise, a Windows Server 2025 system was configured as a domain controller. The roles Active Directory, DNS, and DHCP were installed and set up.

A new domain was created, and the server was promoted to a domain controller. DNS was configured for name resolution, and DHCP was used to assign IP addresses automatically.

Finally, a client computer was added to the domain to verify the functionality of the network services.

3 Theoretische Grundlagen

Active Directory

- Zentrale Datenbank für:
 - Benutzer
 - Computer
 - Gruppen

- Hierarchisch aufgebaut (Domäne, OU, Gesamtstruktur)
 - Verwaltung erfolgt zentral über den **Domain Controller**
 - Hohe Sicherheit (Kerberos, Rechteverwaltung)
-

Domain Controller (DC)

- Server mit AD DS Rolle
- Verwaltet:
 - Anmeldung (Login)
 - Benutzerkonten
 - Richtlinien

Mehrere DCs möglich

DNS (Domain Name System)

- Wandelt **Namen in IP-Adressen**
- Beispiel: server.local → 192.168.1.10

Wichtig für AD:

- AD funktioniert **nicht ohne DNS**

DHCP

- Vergibt automatisch:
 - IP-Adresse
 - Gateway
 - DNS-Server

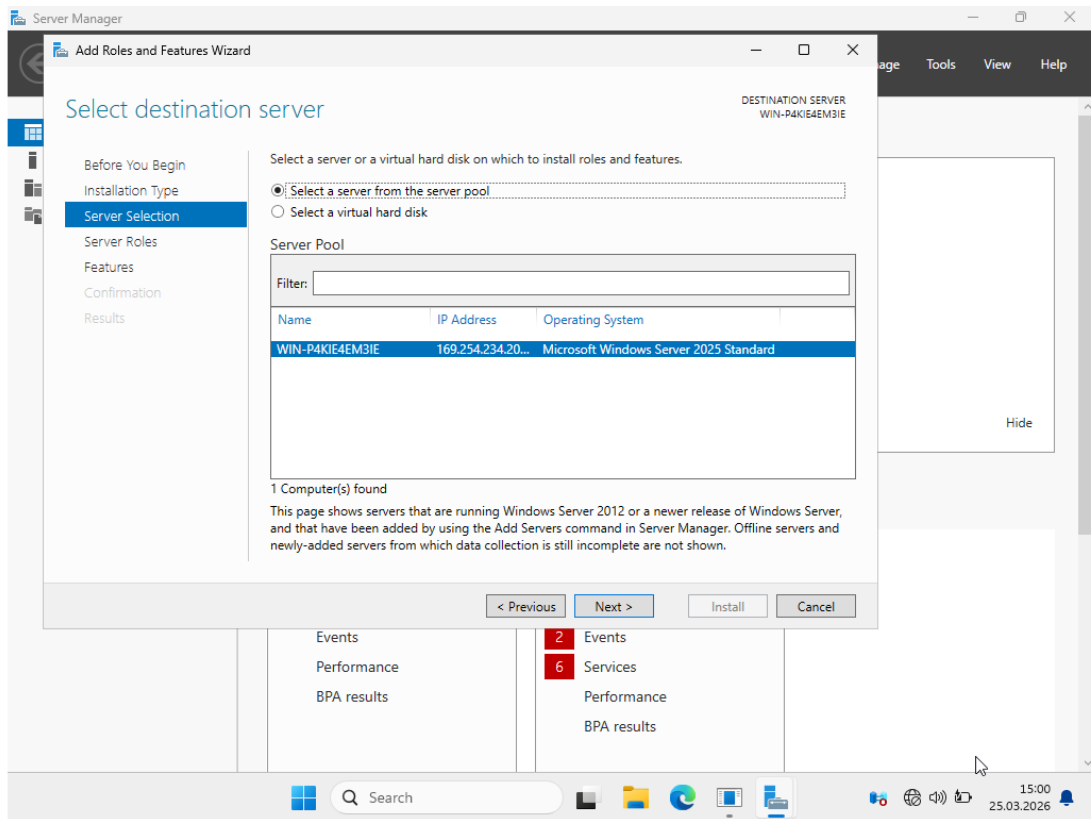
Vorteil:

- Kein manuelles Konfigurieren nötig

4 Übungsdurchführung

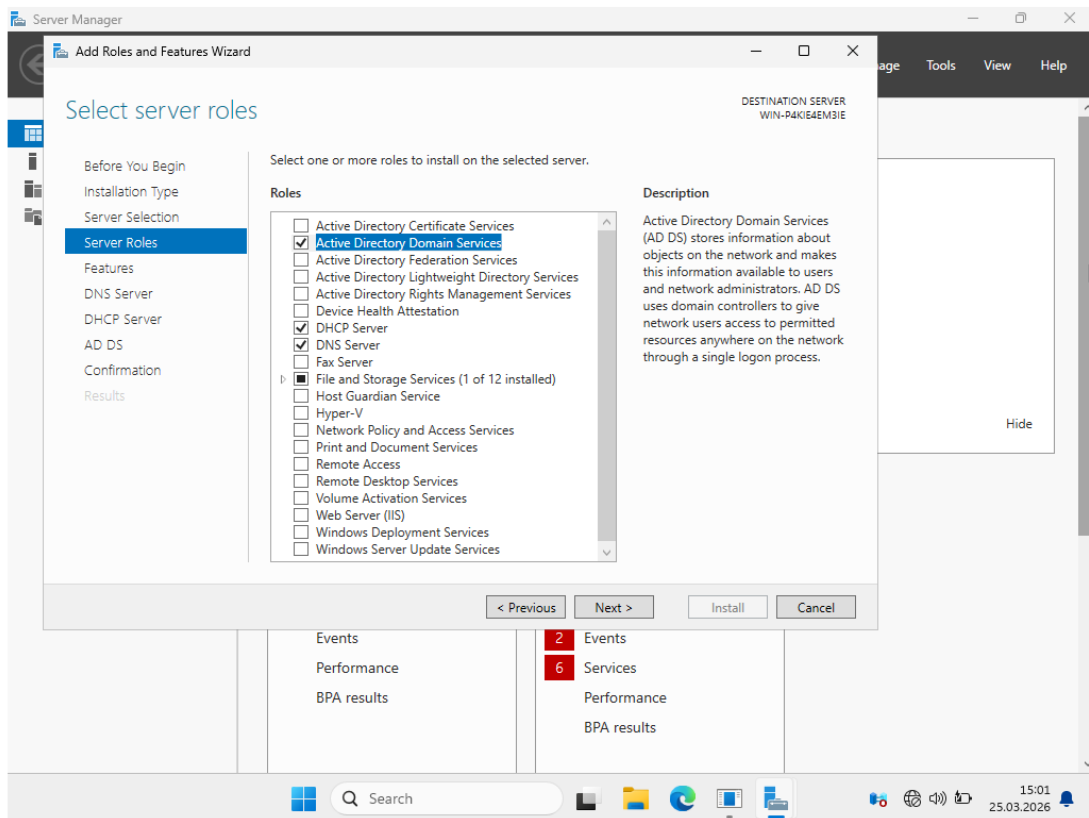
1. Rolleninstallation starten

1.1 Rollen hinzufügen starten



„Select destination server“

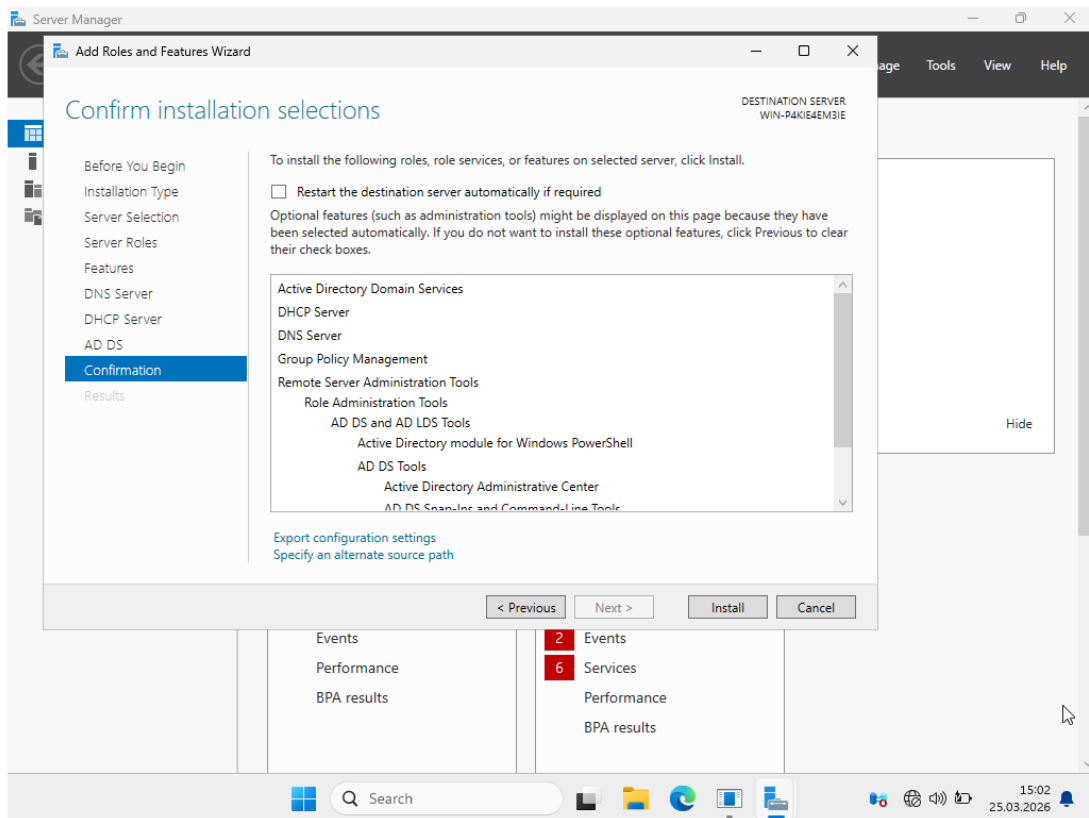
1.2 Serverrollen auswählen



Auswahl von:

- AD DS
- DNS
- DHCP

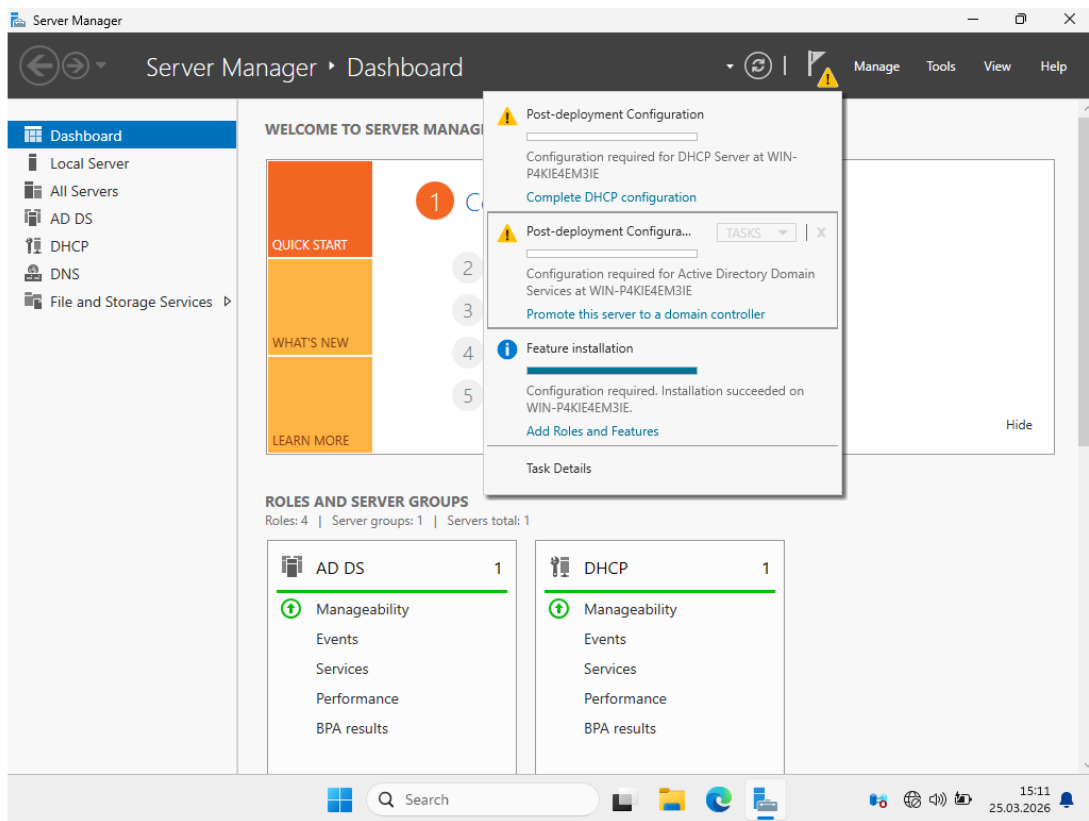
1.3 Installation bestätigen



Confirm installation selections

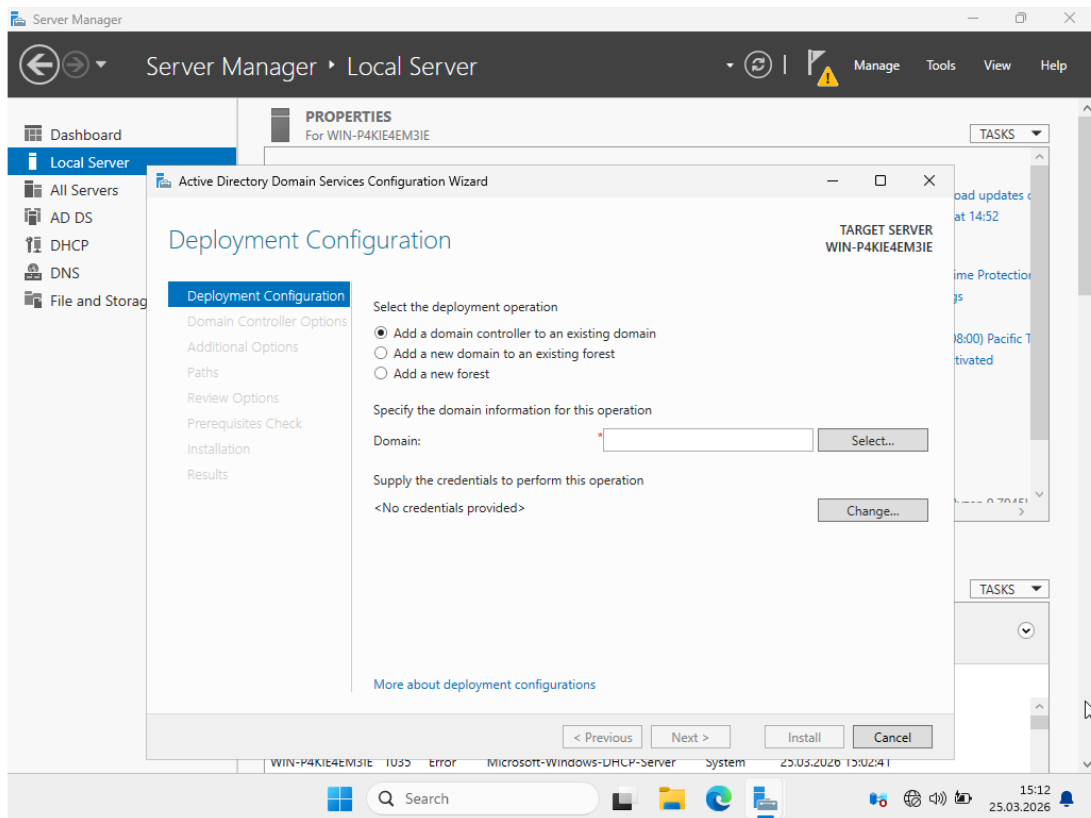
2. Domain Controller einrichten

2.1 Server Manager – Warnung / Promote



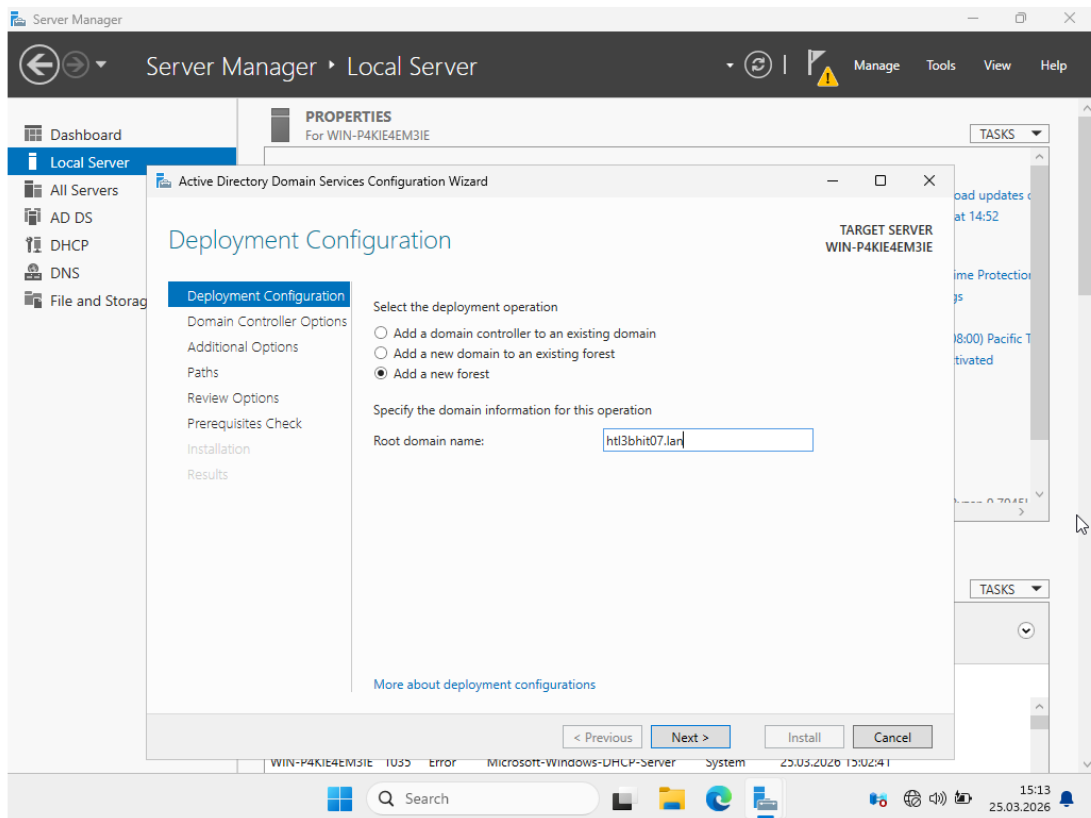
Promote this server to a domain controller

2.2 Deployment Configuration



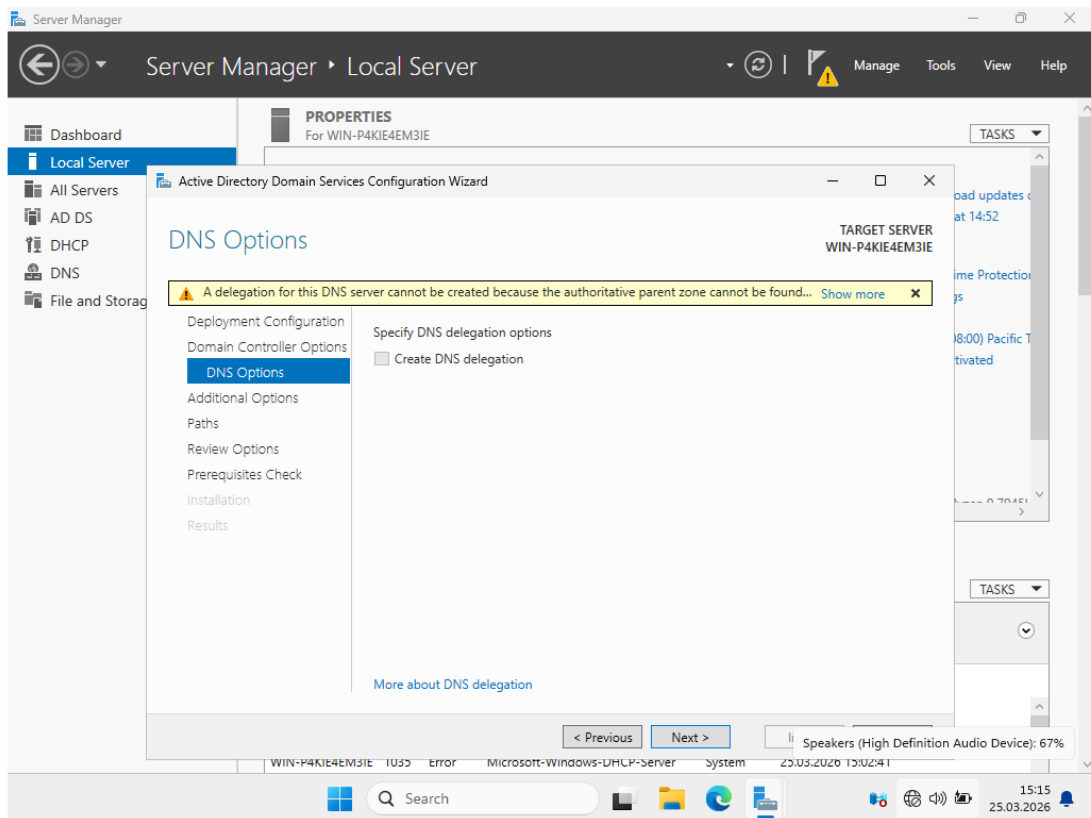
Neue Gesamtstruktur erstellen

2.3 Domänenname eingeben



z. B. htl3bhit04.lan

2.4 DNS Optionen



DNS-Konfiguration (Warnung ist normal!)

3. DNS konfigurieren

3.1 Neue Zone erstellen

New Zone Wizard

Zone Type
The DNS server supports various types of zones and storage.

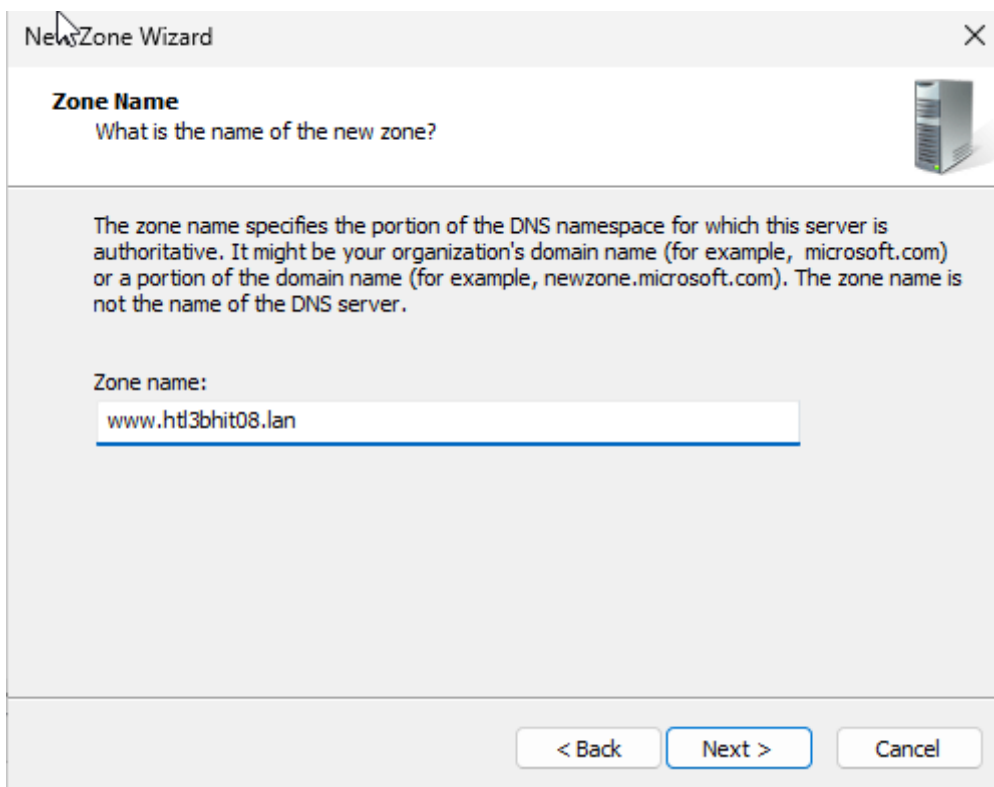
Select the type of zone you want to create:

- ☒ Primary zone
Creates a copy of a zone that can be updated directly on this server.
- ☐ Secondary zone
Creates a copy of a zone that exists on another server. This option helps balance the processing load of primary servers and provides fault tolerance.
- ☐ Stub zone
Creates a copy of a zone containing only Name Server (NS), Start of Authority (SOA), and possibly glue Host (A) records. A server containing a stub zone is not authoritative for that zone.
- ☒ Store the zone in Active Directory (available only if DNS server is a writeable domain controller)

< Back Next > Cancel

Primary Zone auswählen

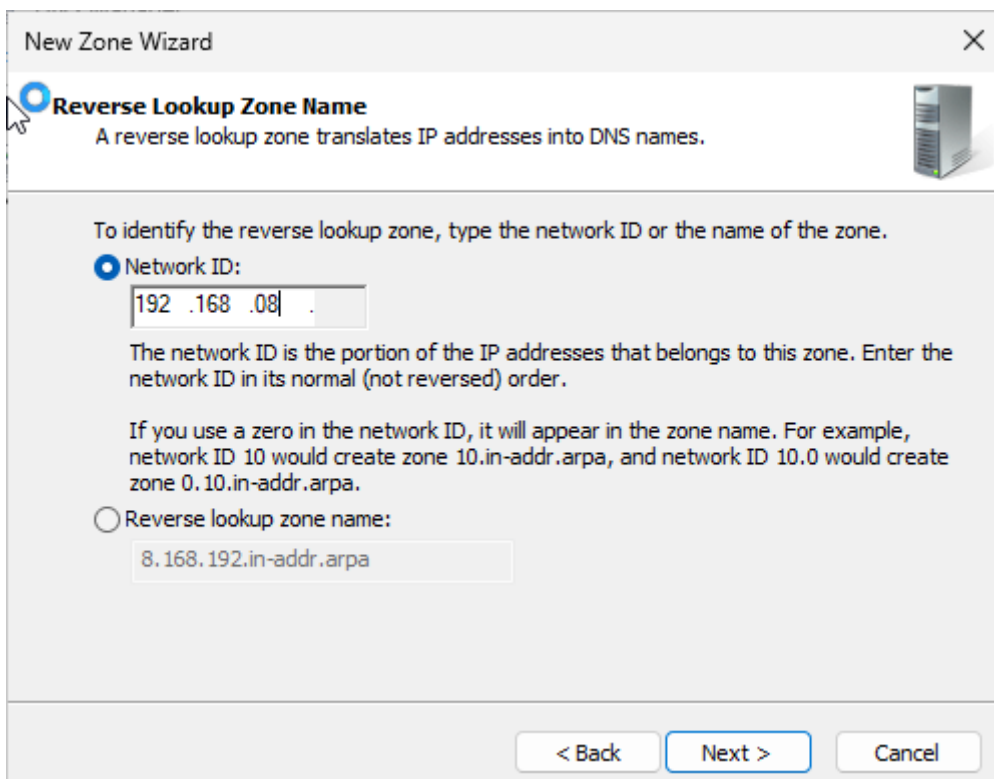
3.2 Zonenname festlegen



z. B.

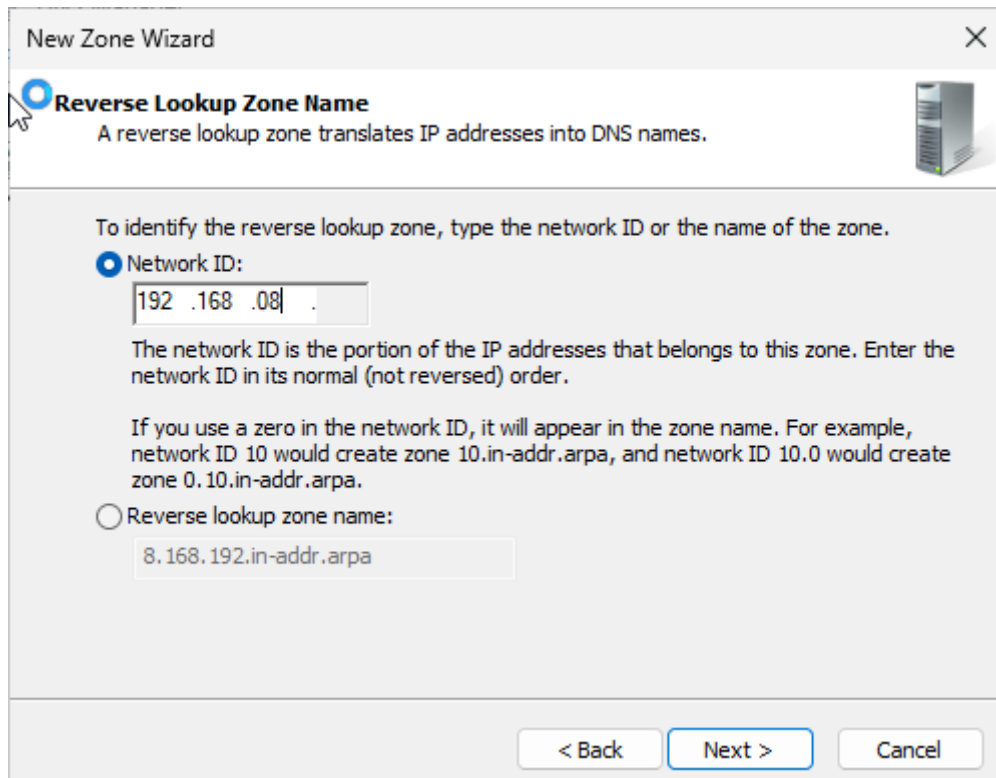
htl3bhit04.lan

3.3 Reverse Lookup Zone



Netzwerk-ID: 192.168.0

3.4 DNS Übersicht

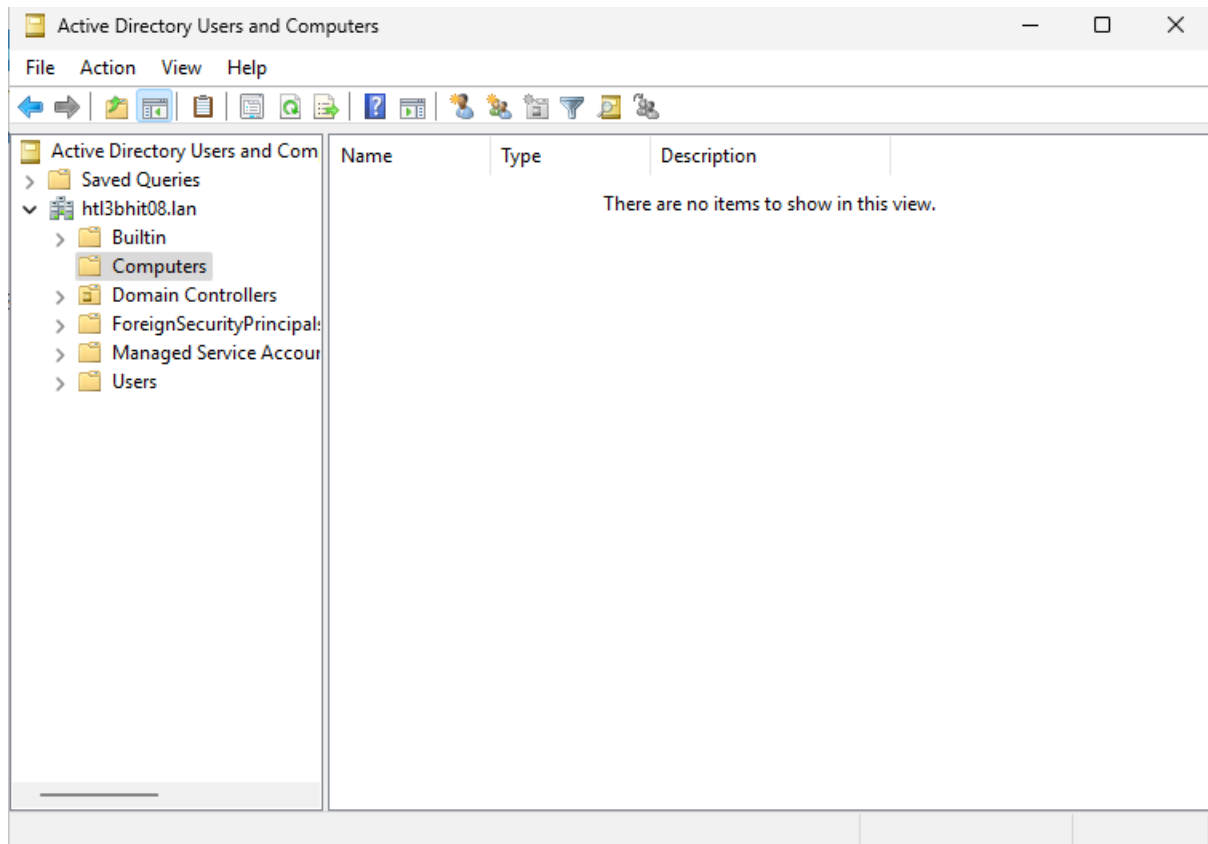


The screenshot shows the 'New Zone Wizard' dialog box with the title bar 'New Zone Wizard' and a close button. The main heading is 'Reverse Lookup Zone Name' with a subtext: 'A reverse lookup zone translates IP addresses into DNS names.' To the right is a server icon. The instruction reads: 'To identify the reverse lookup zone, type the network ID or the name of the zone.' There are two radio buttons: 'Network ID:' (selected) and 'Reverse lookup zone name:'. Under 'Network ID:', there is a text box containing '192 .168 .08' and a dropdown arrow. Below this, explanatory text states: 'The network ID is the portion of the IP addresses that belongs to this zone. Enter the network ID in its normal (not reversed) order.' and 'If you use a zero in the network ID, it will appear in the zone name. For example, network ID 10 would create zone 10.in-addr.arpa, and network ID 10.0 would create zone 0.10.in-addr.arpa.' Under 'Reverse lookup zone name:', there is a text box containing '8.168.192.in-addr.arpa'. At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

DNS Manager mit Zone sichtbar

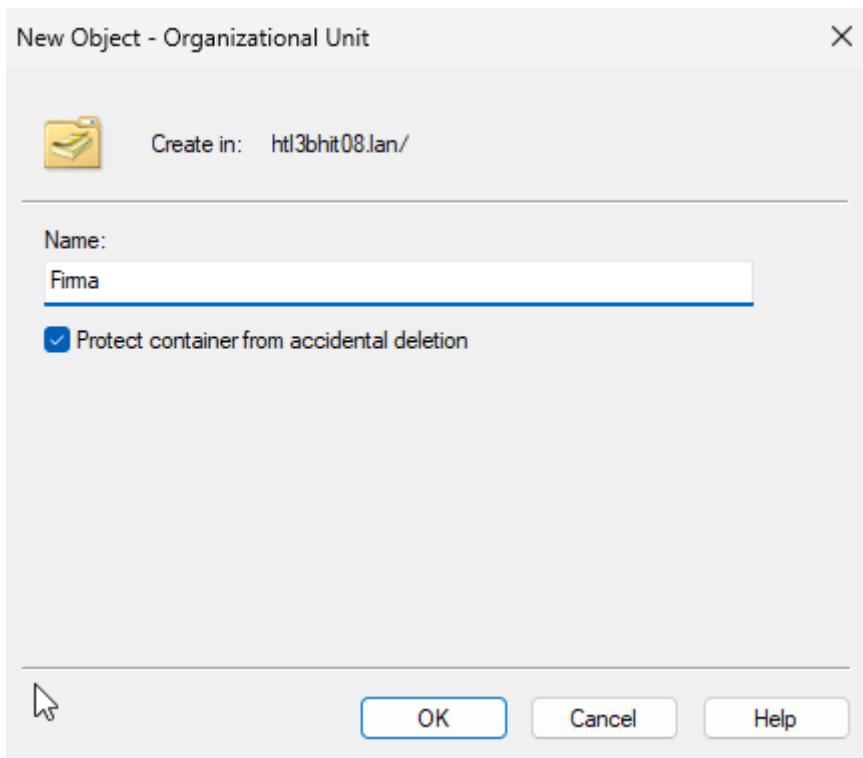
4. Active Directory konfigurieren

4.1 AD Übersicht



Active Directory Users and Computers

4.2 Organisationseinheit erstellen



New Object - Organizational Unit

Create in: htl3bhit08.lan/

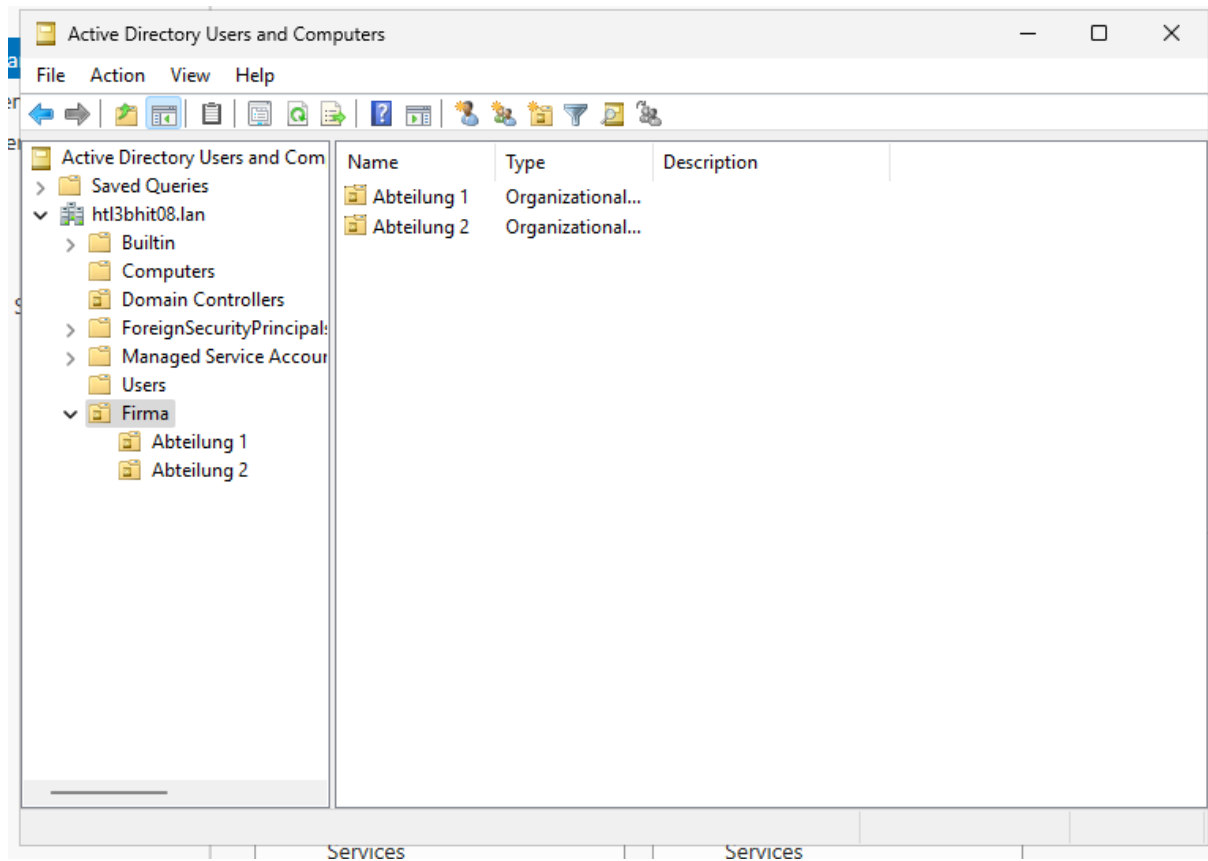
Name:
Firma

☒ Protect container from accidental deletion

OK Cancel Help

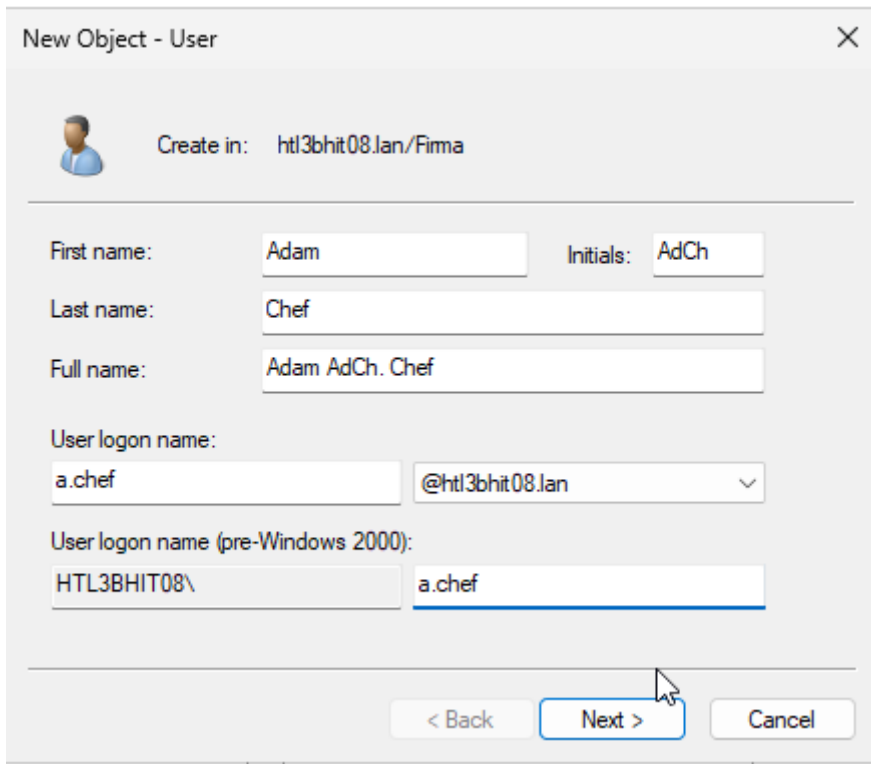
OU „Firma“

4.3 OUs vorhanden



Struktur mit Abteilungen

4.4 Benutzer erstellen



New Object - User

Create in: htl3bhit08.lan/Firma

First name: Adam Initials: AdCh

Last name: Chef

Full name: Adam AdCh. Chef

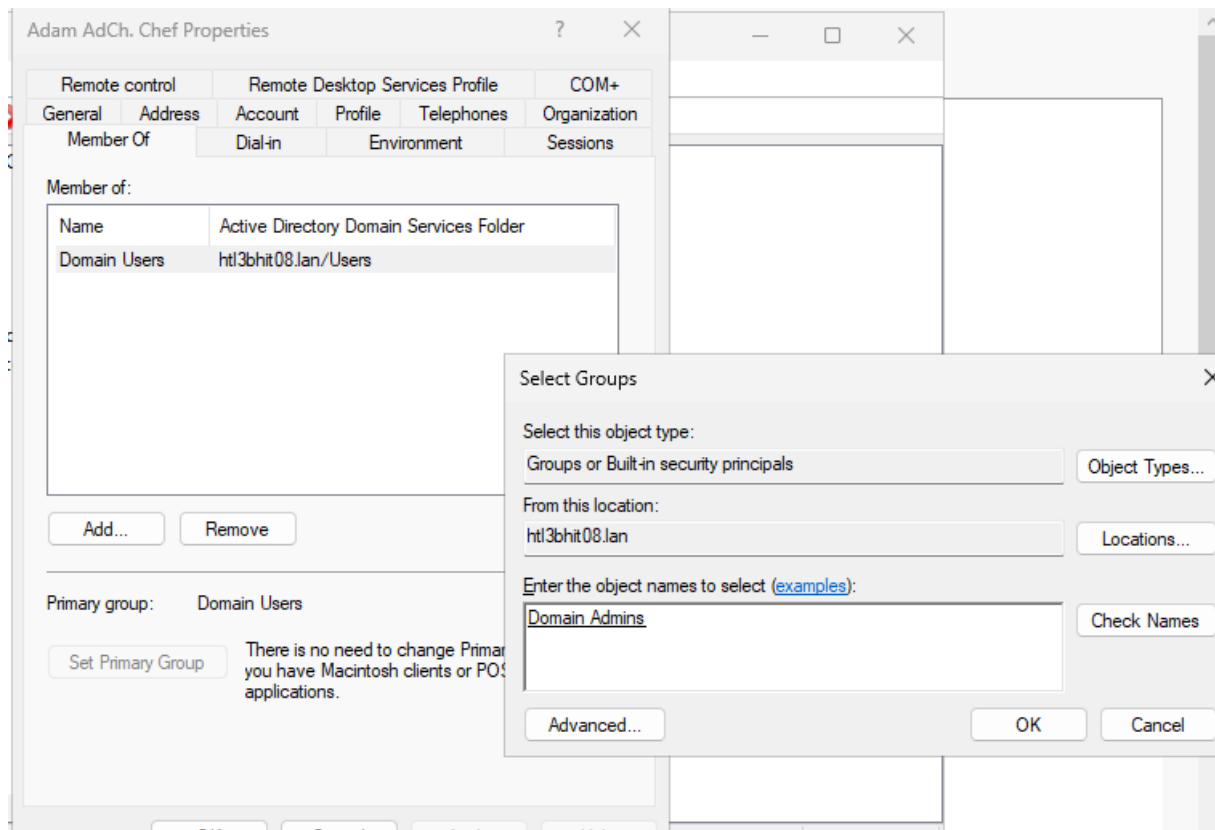
User logon name: a.chef @htl3bhit08.lan

User logon name (pre-Windows 2000): HTL3BHIT08\ a.chef

< Back Next > Cancel

Neuer User (z. B. Adam Chef)

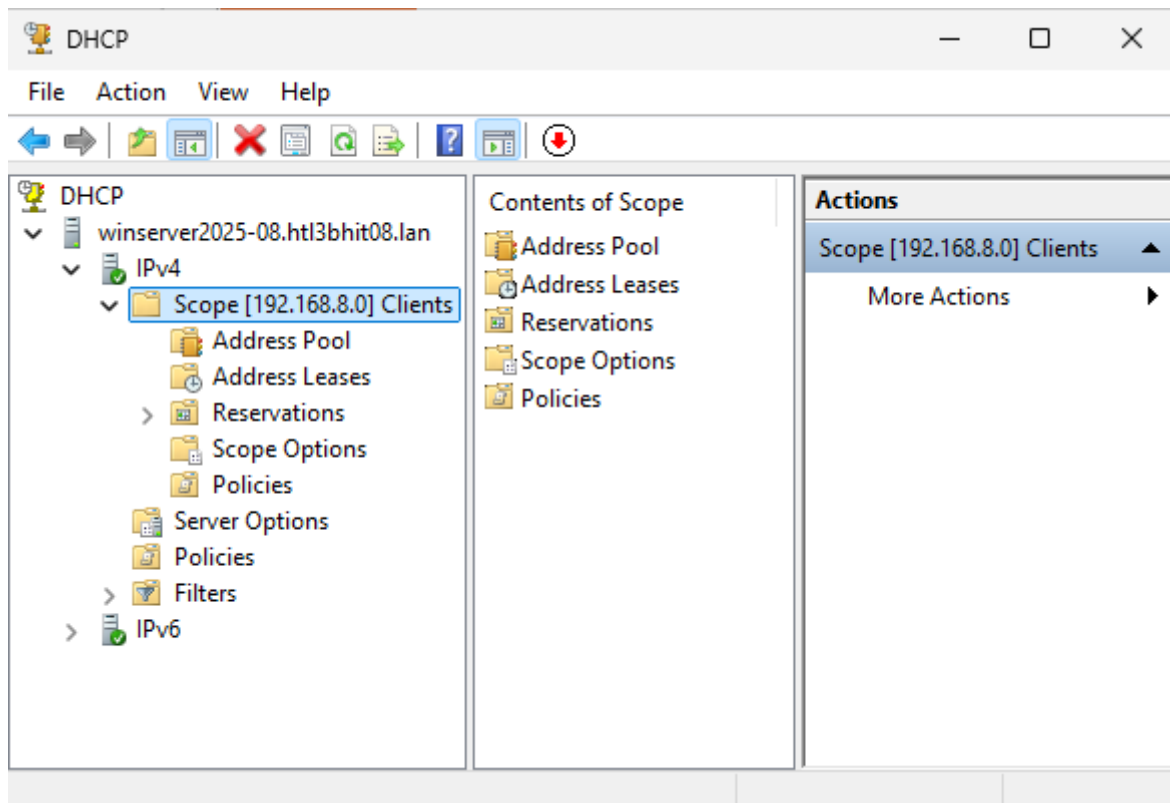
4.5 Benutzer Gruppen zuweisen



Mitglied von Gruppen

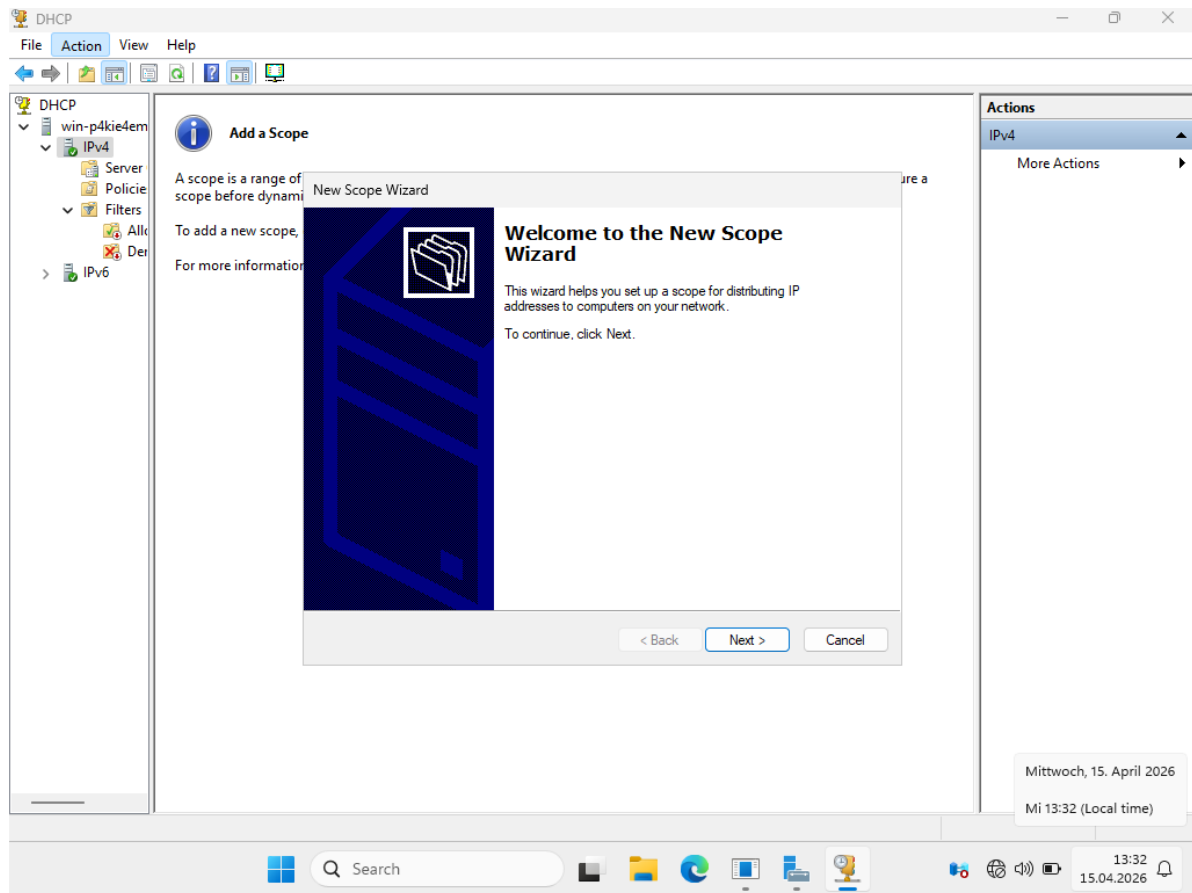
5. DHCP konfigurieren

5.1 DHCP Übersicht



DHCP Manager

5.2 Scope Wizard starten



„New Scope Wizard“

5.3 IP-Bereich definieren

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 192 . 168 . 8 . 100

End IP address: 192 . 168 . 8 . 200

Configuration settings that propagate to DHCP Client

Length: 24

Subnet mask: 255 . 255 . 255 . 0

< Back Next > Cancel

Start: 192.168.0.100

Ende: 192.168.0.200

5.4 Lease Duration

New Scope Wizard

Lease Duration

The lease duration specifies how long a client can use an IP address from this scope.



Lease durations should typically be equal to the average time the computer is connected to the same physical network. For mobile networks that consist mainly of portable computers or dial-up clients, shorter lease durations can be useful. Likewise, for a stable network that consists mainly of desktop computers at fixed locations, longer lease durations are more appropriate.

Set the duration for scope leases when distributed by this server.

Limited to:

Days: Hours: Minutes:

< Back

Next >

Cancel

5.5 Gateway setzen

New Scope Wizard

Router (Default Gateway)

You can specify the routers, or default gateways, to be distributed by this scope.



To add an IP address for a router used by clients, enter the address below.

IP address:

Add

Remove

Up

Down

< Back

Next >

Cancel

z. B.

192.168.0.1

5.6 DHCP aktiviert

Windows-Server [Running] - Oracle VirtualBox

nager

Server Manager ▸ Local Server

DHCP

File Action View Help

← → ↻ ↺ ↻ ?

DHCP	Client IP Address	Name	Actions
- win-p4kie4em3ie.htl3bhit07.lan - IPv4 - Server Options - Scope [192.168.7.0] Scope - Address Pool - Address Leases - Reservations - Scope Options - Policies - Policies - Filters - IPv6	192.168.7.100	DESKT	- Address Leases - More Actions

EVENTS

All events | 43 total

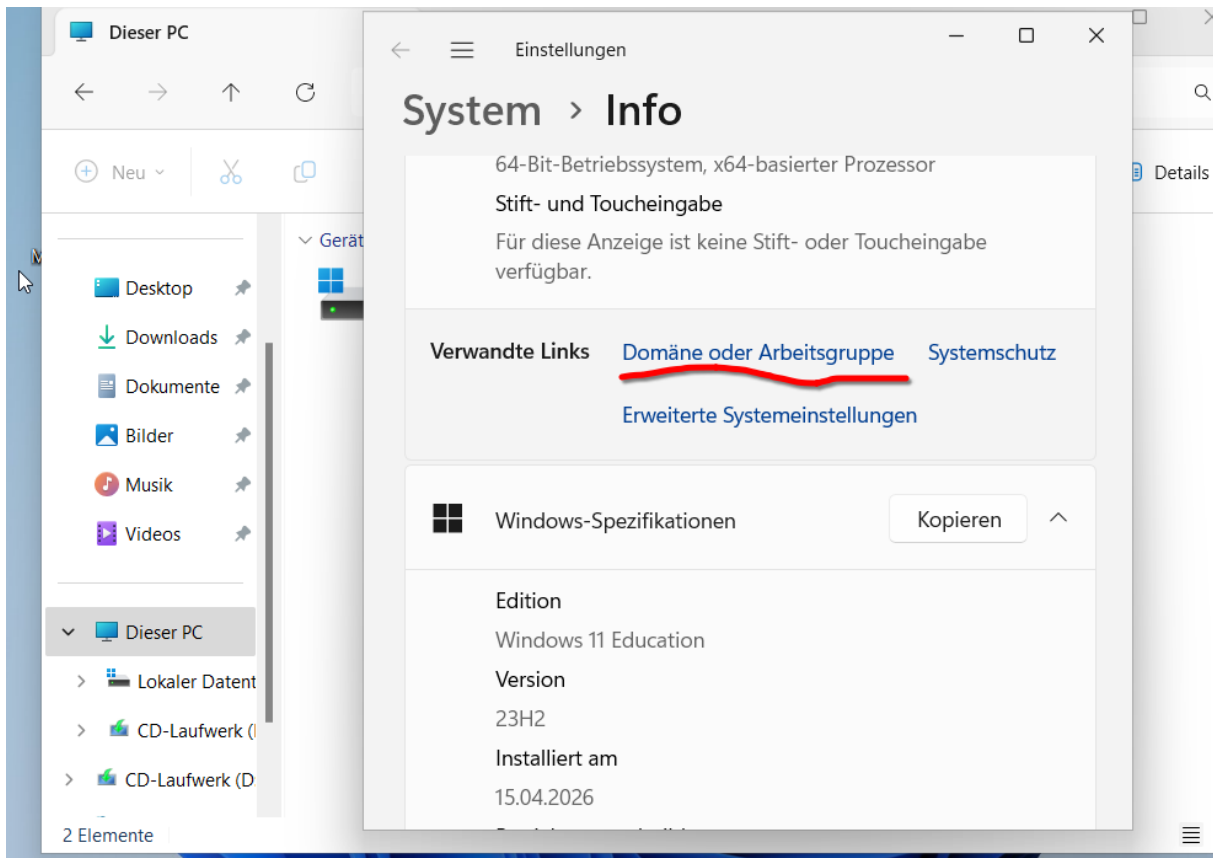
Filter 🔍

Server Name	ID	Severity	Source
WIN-P4KIE4EM3IE	8198	Error	Microsoft-Windows-Security-SPP

Scope aktiv + vergeben

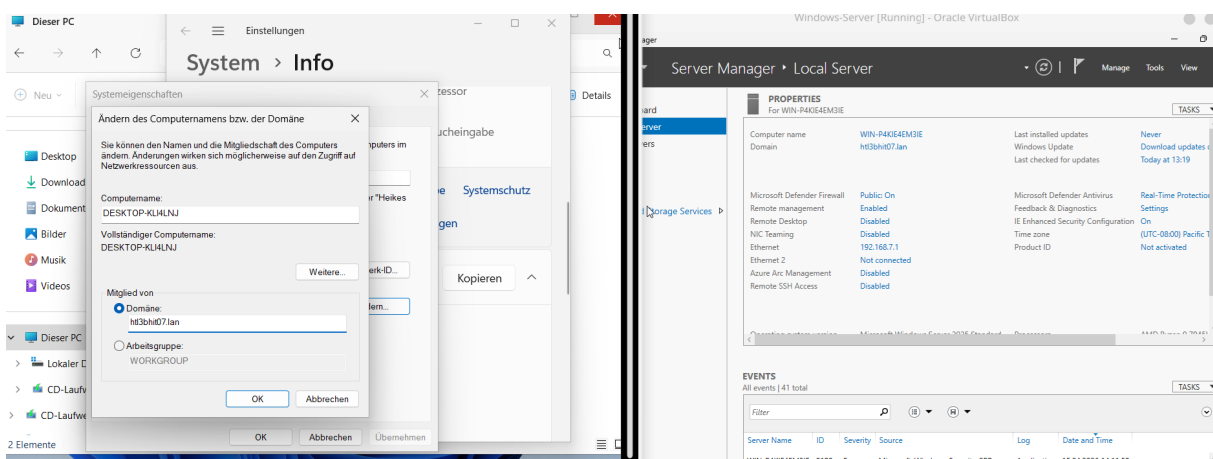
6. Client zur Domäne hinzufügen

6.1 Client Systeminfo



Noch in Arbeitsgruppe

6.2 Domäne beitreten



Änderung zu Domäne

5 Ergebnisse

Client bekommt die Ip nicht

6 Code

7 Kommentar

Ich habe leider keine Ahnung wie ich das Problem beheben kann